

Real Time Information Systems

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Schedule Search For a Real Time System

Figure 1

SOMMAIRE

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Introduction



Getting Started

Execution code of the task τ_1

```
void tau1() {
    long long int i = rand() % 10000;
    long long int j = rand() % 10000;
    i * j;
}

int main() {
    clock_t start, end;
    start = clock();

    tau1();

    end = clock();
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
    printf("Execution time: %f seconds\n", time_taken);

    return 0;
}
```

Ma bite sur ton front ça fait un mal de chien

Retrieving the WCET, Q1, Q2 and Q3 of the task τ_1

```
time_1 = 0
WCET = 0
times = []

security_factor = 1.3

for i in tqdm(range(10000), desc="Calculating WCET"):
    start_time = time.time()
    subprocess.run(["make", "-f", "MakeFile"], stdout=subprocess.DEVNULL,
stderr=subprocess.DEVNULL)
    end_time = time.time()

    time_1 = end_time - start_time

    if time_1 > WCET:
        WCET = time_1
```

```
times.append(time_1)
Q1 = sorted(times)[int(0.25 * len(times))]
Q3 = sorted(times)[int(0.75 * len(times))]
Q2 = sorted(times)[int(0.5 * len(times))]

print("Worst Case Execution Time (WCET) : ", WCET)
print("Secure Execution Time (SET) : ", WCET * security_factor)
print("Q1 : ", Q1)
print("Q2 : ", Q2)
print("Q3 : ", Q3)
```

Coucouibou